Hints: Linear algebra – 5. Bilinear and quadratic forms

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The form has no squares, so first create one. Substitute $x_1 = y_1 + y_2$, $x_2 = y_1 - y_2$, $x_3 = y_3$. Then $x_1x_2 = y_1^2 - y_2^2$ and $x_3(x_1 + x_2) = 2y_1y_3$.

Hint 2. Completing the square gives $Q = (y_1 + y_3)^2 - y_2^2 - y_3^2$, so the signature is $(1, 2)$. Check: the matrix $\frac{1}{2}(J - E)$ has eigenvalues $1, -\frac{1}{2}, -\frac{1}{2}$.

Hint 3. Law of inertia: take two diagonalising bases with p and p' plus signs, and suppose $p > p'$. Compare the span L_+ of the p positive vectors of the first basis with the span $L_\leqslant$ of the $n - p'$ non-positive vectors of the second. Since $p + (n - p') > n$, they share a nonzero x, but then $Q(x) > 0$ and $Q(x) \leqslant 0$. For the example, try $\operatorname{diag}(0, -1)$.

Problem 2

Hint 1. Take $P = tE$. Then $P^TAP = t^2A$, so A and cA have the same minimal polynomial for every $c > 0$. Write $m_{cA}(x) = c^d m_A(x/c)$ and compare coefficients.

Hint 2. So $m_A(x) = x^d$, i.e. A is nilpotent, and the same holds for every P^TAP. Hence $\operatorname{tr}(A\,PP^T) = 0$ for every invertible P. Matrices PP^T (positive definite) span all symmetric matrices, so the symmetric part of A is zero.

Hint 3. A real skew-symmetric matrix is normal, so if it is nilpotent it is 0. Answer: only the zero form $B \equiv 0$.

Problem 3

Hint 1. Write $A = H + K$ with $K = \frac{1}{2}(A - A^T)$ skew-symmetric. Factor out H symmetrically: $A = H^{1/2}(E + K')H^{1/2}$ with $K' = H^{-1/2}KH^{-1/2}$.

Hint 2. K' is again real skew-symmetric, so its eigenvalues are 0 and pairs $\pm ib_j$ with b_j real. Hence $\det(E + K') = \prod(1 + b_j^2) \geqslant 1$.

Problem 4

Hint 1. (Real matrices.) The right-hand side $C^6 + C + E$ is symmetric with eigenvalues $t^6 + t + 1$ for the eigenvalues t of C. Show that $t^6 + t + 1 > 0$ for all real t.

Hint 2. Then $BA(A + E)$ is invertible, so -1 is not an eigenvalue of A. Since $A^3 = -E$, the eigenvalues of A are roots of $x^2 - x + 1$. These are non-real, so they come in conjugate pairs and n is even. Answer: no.

Problem 5

Hint 1. Follow the hint. Check that the hypothesis, together with $a, b, c \in [-1, 1]$, makes all principal minors of $G = \begin{pmatrix} 1 & c & b \\ c & 1 & a \\ b & a & 1 \end{pmatrix}$ non-negative, so $G \succeq 0$ (Sylvester's criterion).

Hint 2. Use the Schur product theorem: the entrywise (Hadamard) product of positive semidefinite matrices is positive semidefinite. Proof idea: $\langle u_i, u_j\rangle\langle w_i, w_j\rangle = \langle u_i \otimes w_i, u_j \otimes w_j\rangle$ is again a Gram matrix.

Hint 3. Hence the n-th Hadamard power $G^{\circ n}$, with entries $1, a^n, b^n, c^n$, is positive semidefinite. Its determinant is $1 + 2(abc)^n - a^{2n} - b^{2n} - c^{2n} \geqslant 0$.